

Demonstrating NeuResonance: Inter-brain Synchronization and Feedback Modalities in Collaboration

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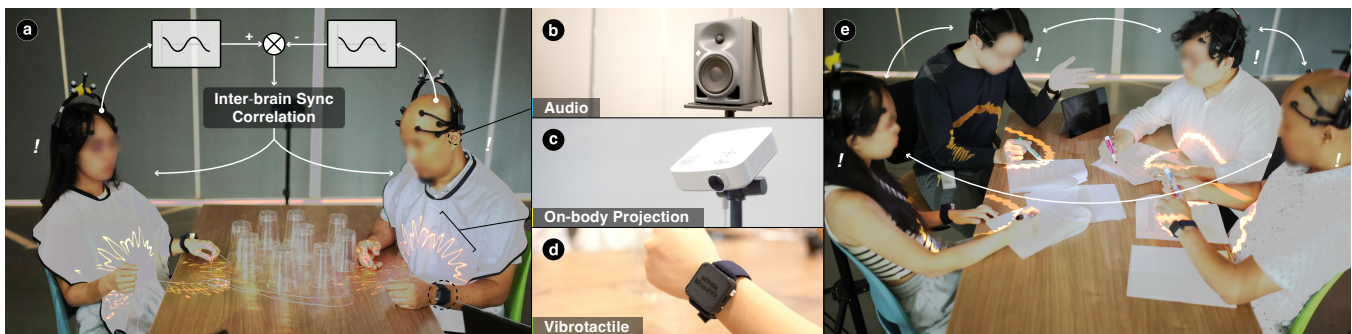


Figure 1: (a) When several individuals work together on a shared task, their brain activities tend to synchronize. This phenomenon, known as *Inter-brain Synchronization (IBS)*, is linked to positive social outcomes. We investigated whether providing feedback on its intensity through (b) auditory, (c) visual, and (d) haptic modalities could further amplify the effect of IBS. (e) This approach could potentially improve team communication and performance, for example, during group brainstorming sessions.

Abstract

When several individuals collaborate on a shared task, their brain activities often synchronize. This phenomenon, known as Inter-brain Synchronization (IBS), is notable for inducing prosocial outcomes such as enhanced interpersonal feelings, including closeness, trust, empathy, and more. The NeuResonance project investigates whether visual, auditory, and haptic feedback of the IBS level can further enhance its intensity, offering design recommendations for feedback systems in IBS. Through this interactive experience, we demonstrate the findings of NeuResonance by showcasing three different types of IBS level feedback, including on-body projection mapping, sonification using chords, and vibration bands attached to the wrist. Using a slider interface visitors can vary the IBS level,

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controlling the feedback exactly how they would have felt if they were the participants of the study.

CCS Concepts

• **Human-centered computing** → **Collaborative interaction**; **Interaction devices**; *Haptic devices*; *Empirical studies in visualization*.

Keywords

Inter-brain Synchronization, Projection mapping, Sonification, Vibrotactile feedback

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1 Introduction

When individuals perform tasks together, their brain activities can synchronize, a phenomenon known as Inter-brain Synchronization

(IBS). This effect has been widely studied in neuroscience, especially in the context of multi-user interactions [19, 30, 31, 36, 38]. Studies have observed shared neural patterns during cooperative problem-solving and synchronized physical activities, including watching the same movie [1], performing coordinated gestures [10], playing musical instruments together [25, 34], engaging in online games [37], holding hands [12], solving puzzles [11], and typing passages collaboratively [27].

Prior research has shown **the association of IBS with prosocial and interpersonal outcomes**. For example, Shiraishi et al. [32] found that IBS is closely linked to behavioral coordination, which enhances ‘we-mode’ processing and fosters a stronger sense of joint agency among participants. Other studies have demonstrated correlations between IBS and feelings of closeness [22], trust [21], empathy [12], pain [12], agreeableness [43], and more. Additionally, the IBS also has the potential to predict collaborative activities by refining the subjective aspects of interactions, including internal experiences and perceptions that individuals bring to and derive from their interactions with others [27].

While studies have demonstrated the correlation between IBS and positive prosocial outcomes, research on methods to **actively enhance IBS remains unexplored**, possibly because IBS is still an emerging area in HCI [15, 30]. Semertzidis et. al [30] developed a wearable brain-to-brain system to augment IBS through magnetic brain stimulation. Their study demonstrated that enhancing IBS fostered group empathy, group performance, and relational interactions. While these findings highlight the potential of IBS in facilitating better prosocial experiences, little is known about optimal feedback design and efficacy for enhancing IBS through external feedback that is widely accessible to end-users.

To address this gap in the literature, we propose a study aimed at answering the question: **How can external feedback be used to actively enhance IBS between individuals?** Our findings would serve as an initial step toward achieving unique teamwork and creating synergies beyond what individuals can achieve alone. For instance, enhancing IBS could be beneficial in educational settings where students collaborate on complex problems, in corporate environments to improve teamwork and communication, or in therapeutic contexts to strengthen the bond and sympathy between therapist and patient [24].

Motivated by prior works in IBS, feedback systems, and how external stimuli influence brainwaves, the project NeuResonance [9] aims to identify the most effective modality and feedback design for enhancing IBS in visual, auditory, and haptic modality, ultimately testing a hypothesis that providing external feedback during collaboration can enhance IBS intensity between participants.

2 Implementation

Our feedback system includes EEG measurement, a motion capture system, data analysis that encompasses preprocessing and IBS correlation estimation, and feedback systems. The details are provided below.

2.1 Dual-EEG data acquisition

The EEG data was collected using two Emotiv EPOC X wireless EEG headsets, each equipped with 14 channels. The Emotiv EPOC

X has been validated as a reliable, research-grade system for EEG studies [40], and has been widely adopted in various brain activity analyses, including research on IBS [7, 8] and other areas of neuroimaging [3, 39]. The headset’s lightweight and wireless design makes it well-suited for the motor collaborative tasks we developed. From our pilot studies, we were able to observe the IBS phenomenon with Emotiv EPOC X and validate it through 6 different sets of headsets. Both headsets were connected to a single computer, with raw EEG data streams captured through an open-source Lab Streaming Layer (LSL) [20] protocol. All relevant data was collected across all 14 channels, using a sampling rate of 256 Hz.

2.2 Real-time EEG Analysis

We implemented a real-time EEG analysis to compute IBS circular correlation values, which were used to adjust the feedback. Our goal was to analyze participants’ brain activity as they interacted with the system, utilizing two 14-channel raw EEG data streams. EEG data streams are fed into Python code utilizing the MNE library [13] and the Hyperscanning Python Pipeline for inter-brain connectivity analysis (HyPyP) [4]. HyPyP provides various inter-brain connectivity measures with both automated and manual data preprocessing. This toolbox, widely used in IBS research, allows us to compute inter-brain connectivity [2, 28, 35, 37], providing insights into neural mechanisms that underlie interactive experiences and how they facilitate communication and collaboration in mixed reality settings.

2.2.1 Preprocessing and Motion Artifact Removal. Preprocessing ensures that the EEG data is clean, interpretable, and reliable, improving the overall quality and accuracy of the subsequent analysis. In prior studies, the selection of epoch lengths of 1 to 1.5 seconds for inter-brain measure estimation is a common practice. This duration is often preferred as it provides a balance between accurately capturing the transient changes in neural activity and ensuring the dependability of the data obtained [8, 10, 27]. However, recent studies have shown that shorter windows lead to unreliable estimations, particularly in cases of noisy signals [5]. Real-time EEG raw data from two participants were recorded into two separate buffers of a sliding window of 3 seconds with a 1.5 seconds hop. The signals were then filtered with a band-pass filter for frequencies between 1 Hz and 48 Hz. The filtered data was used to calculate the IBS circular correlation value using HyPyP. In parallel, to ensure efficient real-time data preprocessing, we used Vicon motion capture to analyze the participants’ head motion, as described in section 2.2.1, to reject large and fast motions. While the system updates IBS every ~ 100 ms, the preprocessing algorithm is highly optimized with each process, with a total execution time of ~ 40 – 60 ms for IBS circular correlation and ~ 2 ms for Vicon based motion artifact classification.

Considering the gestural nature of the collaborative tasks, we also augmented the EEG headset with the Vicon motion capture (MoCap) markers to track both linear and angular fast and large motion and reject them from our calculations. The Vicon MoCap system is submillimeter accurate [23], sampled at 100 Hz, and gives both linear and angular position.

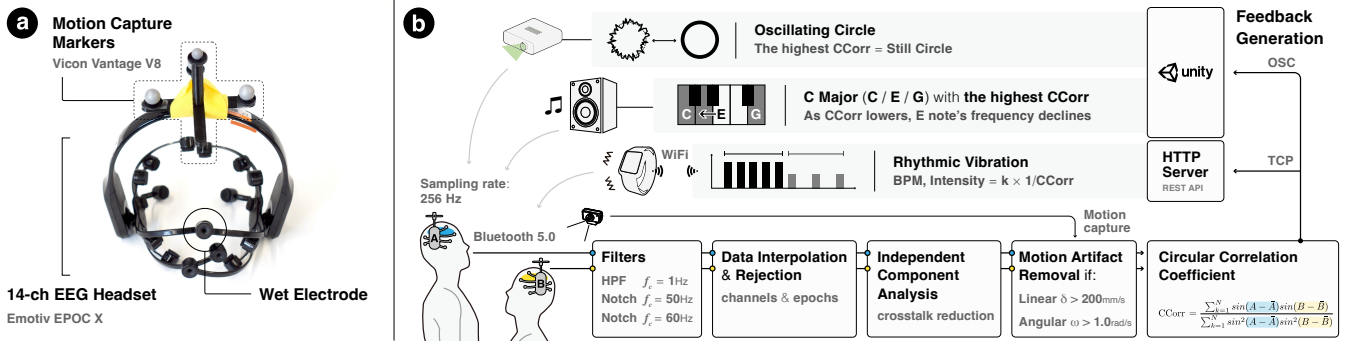


Figure 2: (a) Emotiv EPOC X EEG headset with mocap markers (b) Block diagram of the proposed feedback system

2.2.2 Measuring Inter-brain Synchronization. After pre-processing, the system calculated the level of IBS between two participants. Although Phase Lock Value (PLV) and Phase Lock Index (PLI) were widely used in previous research to measure IBS [10, 18, 26, 29], however, Burgess et al. [6] suggested that CCorr shows more robustness to coincidental synchrony. The circular correlation coefficient (CCorr) measures IBS by assessing the alignment of phase angles in brain signals between individuals, capturing how neural oscillations are synchronized during shared activities. High covariance and CCorr values near 1 indicate strong synchrony, while low covariance and CCorr values approaching 0 suggest minimal synchrony [16].

CCorr for two corresponding EEG channels is:

$$\text{CCorr}_{A,B} = \frac{\sum_{k=1}^N \sin(A - \bar{A}) * \sin(B - \bar{B})}{\sum_{k=1}^N \sin^2(A - \bar{A}) * \sin^2(B - \bar{B})} \quad (1)$$

where A and B are the phases of two EEG signals, and $\bar{A}$ and $\bar{B}$ are the average phases of respective signals.

For each buffer of 3 seconds, the system calculated channel-to-channel CCorr from two participants. From the 14 CCorr values derived from each channel, the average of the five highest CCorr values is used as the IBS metric. To mitigate bias introduced by averaging multiple correlation coefficients, the top 5 pairwise CCorr values were transformed using Fisher's z-transformation [30]. The z-transformed values were averaged, and this average was subsequently converted back using the inverse Fisher's z function. Fisher's z-transformation is defined as follows:

$$z = \frac{1}{2} \log_e \left(\frac{1+r}{1-r} \right) \quad (2)$$

With the buffers of 3 seconds sliding window with a 50% overlap [8, 10, 27], the system outputs a CCorr value every 1.5 seconds. Based on our pilot study of 16 sessions, we observed that the IBS metric was approximately 14% higher during collaborative tasks compared to non-collaborative situations.

2.3 Feedback System

Through the proposed IBS metric, the average of the top five highly correlated CCorr values from 14 channels are updated every 1.5 seconds. The metric is sent via OSC [42] and WIFI to each feedback platform. Three different types of modalities, audio, visual, and

haptic, are designed as the feedback. The level of feedback was divided into five subdivisions according to the IBS metric.

2.3.1 Visual Feedback: Projection Mapping. The proposed application creates an abstract visualization of an oscillating circle using Unity that is controlled by the IBS value. The oscillation has a negative relationship with the IBS value; as the IBS value increases, the oscillation decreases (Fig. 3-1). Consequently, a perfect IBS value would result in a still circle (Fig. 3-4), while a low IBS value would correspond to increased oscillation. To project the visual, we used a pair of LG PF50KA portable projectors with 600 ANSI lumens brightness. Our on-body projection mapping approach makes the system expandable to the multiuser feedback in a further group collaboration scenario.

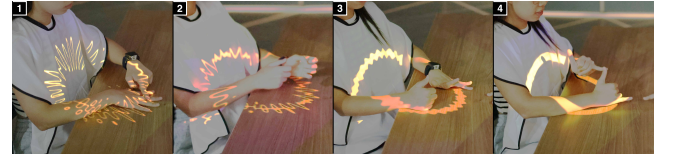


Figure 3: Different levels of IBS are illustrated by a ring-shaped visual representation. The amplitude of the waveform corresponds to the IBS intensity, where the sync levels are: (1) the lowest, (2) low, (3) high, and (4) the highest.

2.3.2 Auditory Feedback: Frequency Shift. In our approach, the IBS value is mapped to the frequency of a middle note in a musical chord. A harmonic musical chord, a group of three or more notes played simultaneously, is used to guide the participants to perceive the shift of the mid-notes frequency perceivably. Especially, the harmonic series chord is a sequence of frequencies that are integer multiples of a fundamental. The utilized Major triad chord has three notes with a 4:5:6 frequency ratio. Among the three notes, the first (root) note and the third (perfect fifth) note have a 2:3 ratio, which is a simple, low-integer ratio that creates a highly stable and consonant sound. Our auditory system processes sound in ways that make certain intervals feel more “stable” or “pleasant [41].”

To induce this effect, the middle note's frequency shifts from the major third to a lower frequency based on the IBS level. One level degradation of IBS corresponds to a 5% decline of the middle note's frequency distorting the ratio of harmonic series more noticeably.

While the just-noticeable difference (JND) is about 3 – 5 Hz around 500 Hz – 600 Hz, our frequency shift occurs between 547 Hz – 659 Hz over 5 different levels [17]. During the feedback, our Unity code synthesizes the corresponding notes as a chord and plays them every 1.5 seconds, synchronized with the IBS metric update. The audio is created by the 30 watt stereo speakers in front of the participants.

2.3.3 Haptic Feedback: Vibration Band. We utilized a haptic band as shown in Fig. 1d that builds upon [33] to provide vibrotactile feedback in the form of repeating beats where faster rate in the form of beats per minute (BPM) and stronger intensities indicating higher dissonance while slower BPM and lower intensities indicating higher synchrony. The feedback was divided into five subdivisions with increment following [14] characterization of the haptic stimulus-sensation relationship.

We conducted a pilot study (N=5) in which participants provided a thumbs-up or thumbs-down signal to indicate whether they perceived the vibrations as faster and more intense or slower and less intense, respectively. Participants were instructed to disregard instances where they were unsure of their perception. We systematically varied BPM and intensity values and observed participants' ability to discern changes. Initially, we tested 20 intervals, informed by [14], but refined this to 10 intervals based on experimental findings. To ensure consistency across modalities and enhance perceptual distinction, we decided the intervals to five in the final study.

The haptic band (hBand v2 [33]) hardware is manufactured by vAlolin. It is powered by an ESP8266 microcontroller with WiFi connectivity pulse-width modulation (PWM) output connected to four eccentric rotating mass (ERM) vibration motors (VC1030B028F). The device is compactly integrated into a wearable watch-like 3D printed form factor weighing 200 g, powered by a LiPO battery. The devices accept REST API to control the vibration patterns as described in [33].

3 Envisioned Interaction

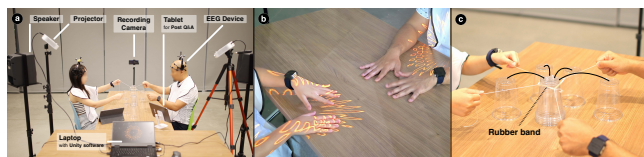


Figure 4: (a) Interactivity setup (b) Hand Mimicry session (c) Cup & String session

Visitors will have the opportunity to see, hear, and feel the level of IBS at the conference venue. We will replicate the study setup using multiple projectors, speakers, and vibration bands to deliver modality feedback to visitors and experimenters (Figure 4a). Many visitors can visit our booth and try out how each modality of another person's EEG signal translated into feedback modalities looked and felt like. We will demonstrate how feedback is provided when there is good synchronization between two collaborators, allowing users to experience how it feels to sense the other person's state through our input modality in each method.

Our demo provides a fast-forward version of the experience inviting a larger audience. Considering the complexity of EEG setup, optionally visitors can experience the feedback through the recorded EEG data for better concept communication in the limited time. Using a slider interface visitors can vary the IBS level, controlling the feedback (visual from the projectors, audio from the speakers, and vibration from the haptic band) exactly how they would have felt if they were the participants of the study.

Additionally, the cup and string task will also be present, and visitors can try the task with a stranger or with one of their friends (Figure 4b, 4c).

4 Conclusions

This study investigated how Visual, Auditory, and Haptic feedback influence Inter-brain Synchronization (IBS) and provided design recommendations for feedback systems aimed at enhancing IBS. More details on the experiment, result, and discussion are available in [9]. By employing EEG to measure IBS metrics, our results reveal that auditory feedback had the most substantial and consistent impact on improving IBS. Haptic feedback also positively affected IBS intensity, whereas Visual feedback, delivered through on-body projection mapping, did not show significant improvements compared to no feedback. These findings validate our hypothesis that feedback generally enhances IBS intensity and highlight opportunities for further improvement through task-adaptive design, personalization, and multimodal approaches.

While traditional EEG and IBS user studies can be time-consuming to set up and cumbersome to clean, our demonstration allows more people to quickly experience the effects of feedback on IBS. By significantly reducing preparation time, we are able to let more visitors see and feel the feedback systems in action, aligning with the authors' objective of making the experience accessible to a wider audience. This approach not only enhances user engagement but also provides crucial insights for developing effective feedback systems that can boost IBS and enhance user experiences.

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